

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R

PRACTICAL 13

AIM: Identifying and handling duplicates using `distinct()` (R studio).

PURVI ROSHAN KARKERA S086

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```
RPROG - RStudio
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Source
R 4.5.2 -- RPROG

> # =====
> # R Scripts: Identifying and Handling Duplicates
> # Functions: distinct() from the dplyr package
> # Dataset: arxiv_ai.csv (User Uploaded)
> # =====
> library(dplyr)
> # =====
> # 1. SETUP: Create a Dataset with Intentional Duplicates
> # =====
> # Load original dataset
> df <- read.csv("arxiv_ai.csv", na.strings = c("", "NA"))
>
> # Create PaperID for easier tracking
> df <- df %>%
+   mutate(
+     PaperID = row_number(),
+     Title = title,
+     Category = categories
+   )
>
> # Create intentional duplicates:
> # - Duplicate row 1 and 6
> # - Duplicate row 2 and 3
> # - Category may be same but titles different in some rows
> duplicate_df <- df[c(1, 2, 3, 4, 5, 1, 2), c("PaperID", "Title", "Category")]
>
> print("---- 1. Original Dataset With Intentional Duplicates (7 rows) ----")
[1] "---- 1. Original Dataset With Intentional Duplicates (7 rows) ----"
> print(duplicate_df)
  PaperID
1        1
2        2
3        3
4        4
5        5
6        1
7        2

                                Title
1 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
2 Dynamic Backtracking
3 An Empirical Analysis of Search in GSAT
4 The Difficulties of Learning Logic Programs with Cut
5 Software Agents: Completing Patterns and Constructing User Interfaces
6.1.1 Dynamic Backtracking
7.2.1 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems

Environment History Connections Git Tutorial
R Import Dataset 285 MB
Global Environment
clean_exact 5 obs. of 3 variables
combined_data 10000 obs. of 2 variables
df 10000 obs. of 15 variables
df_clean 10000 obs. of 4 variables
df_part1 5000 obs. of 2 variables
df_part2 5000 obs. of 2 variables
duplicate_df 7 obs. of 3 variables
duplicates_report 2 obs. of 4 variables
job_title 3000 obs. of 18 variables
long_df 20000 obs. of 4 variables
unique_titles 5 obs. of 3 variables

Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Name Size Modified
.gitignore 44 B Dec 1, 2025, 11:03 AM
RData 86.3 KB Nov 24, 2025, 12:31 PM
Rhistory 457 B Dec 1, 2025, 11:12 AM
ALImpact_on_Jobs_2030.csv 296.4 KB Nov 24, 2025, 11:43 AM
analysis.R
prac.6.7.8.9.10
prac.6.7.8.9&10.r prog link.R
prac.pdfs
README.md 23 B Nov 24, 2025, 6:03 PM
RPROG.Rproj 218 B Dec 8, 2025, 10:30 AM
arxiv_ai.csv 13.3 MB Dec 8, 2025, 10:58 AM
```

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```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to File/function Addins
Source
R4.5.2 - RPROG.R
> print("--- 1. Original Dataset With Intentional Duplicates (7 rows) ---")
[1] "--- 1. Original Dataset With Intentional Duplicates (7 rows) ---"
> print(duplicate_df)
  PaperID
1       1
2       2
3       3
4       4
5       5
1.1     1
2.1     2

  Title
1 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
2 The Difficulties of Learning Logic Programs with Cut
3 An Empirical Analysis of Search in GSAT
4 Software Agents: Completing Patterns and Constructing User Interfaces
5 Dynamic Backtracking
1.1 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
2.1 Dynamic Backtracking
  Category
1 ['cs.AI']
2 ['cs.AI']
3 ['cs.AI']
4 ['cs.AI']
5 ['cs.AI']
1.1 ['cs.AI']
2.1 ['cs.AI']

> # 2. IDENTIFYING DUPLICATES (Before Removing Them)
> #
> duplicates_report <- duplicate_df %>%
+   group_by(PaperID, Title, Category) %>%
+   count() %>%
+   filter(n > 1) # Keep only duplicates
>
> print("--- 2. Identification Report (Rows that appear more than once) ---")
[1] "--- 2. Identification Report (Rows that appear more than once) ---"
> print(duplicates_report)
# A tibble: 2 x 4
# Groups:   PaperID, Title, Category [2]
  PaperID Title Category n
  <int> <chr> <chr> <int>
1 1 Dynamic Backtracking ['cs.AI'] 2
2 2 A Market-Oriented Programming Environment and its Application to Distributed Multicommod... ['cs.AI'] 2

> # 3. HANDLING DUPLICATES: Exact Matches
```

Environment History Connections Git Tutorial

Object	Size	Variables
clean_exact	5 obs.	of 3 variables
combined_data	10000 obs.	of 2 variables
df	10000 obs.	of 15 variables
df_clean	10000 obs.	of 4 variables
df_part1	5000 obs.	of 2 variables
df_part2	5000 obs.	of 2 variables
duplicate_df	7 obs.	of 3 variables
duplicates_report	2 obs.	of 4 variables
job_title	3000 obs.	of 18 variables
long_df	20000 obs.	of 4 variables
unique_titles	5 obs.	of 3 variables

Files Plots Packages Help Viewer Presentation

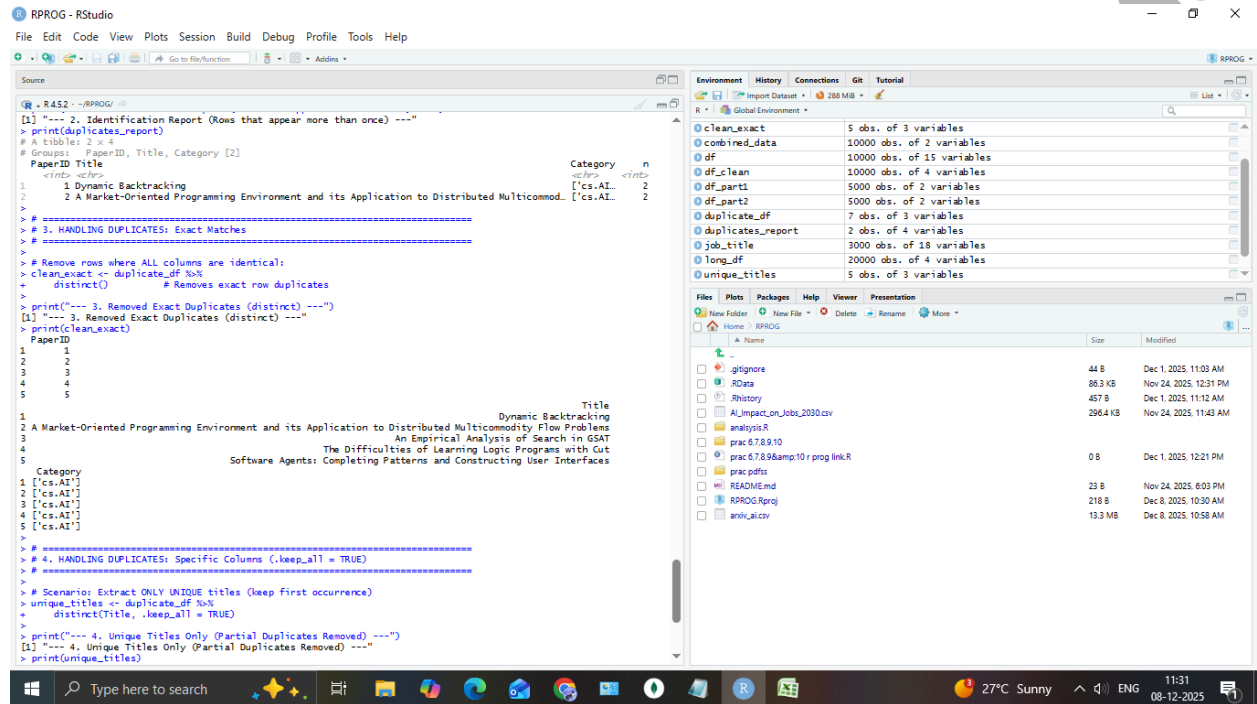
Name	Size	Modified
.gitignore	44 B	Dec 1, 2025, 11:03 AM
.RData	86.3 KB	Nov 24, 2025, 12:31 PM
.Rhistory	457 B	Dec 1, 2025, 11:12 AM
AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac.6.7.8.9.10		
prac.6.7.8.9&10.r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac.pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
analysis.csv	13.3 MB	Dec 8, 2025, 10:58 AM

Windows Taskbar: Type here to search, 27°C Sunny, 11:31, 08-12-2025

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RPROG - RStudio

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Source

```
R 4.5.2 - ~/RPROG/
PaperID
1 1
2 2
3 3
4 4
5 5

Title
1 Dynamic Backtracking
2 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
3 An Empirical Analysis of Search in GSAT
4 The Difficulties of Learning Logic Programs with Cut
5 Software Agents: Completing Patterns and Constructing User Interfaces

Category
1 ['cs.AI']
2 ['cs.AI']
3 ['cs.AI']
4 ['cs.AI']
5 ['cs.AI']

> # =====
> # 4. HANDLING DUPLICATES: Specific Columns (.keep_all = TRUE)
> # =====
> # Scenario: Extract ONLY UNIQUE titles (keep first occurrence)
> unique_titles <- duplicate_df %>%
+   distinct(Title, .keep_all = TRUE)
>
> print("---- 4. Unique Titles Only (Partial Duplicates Removed) ----")
[1] "---- 4. Unique Titles Only (Partial Duplicates Removed) ----"
> print(unique_titles)
PaperID
1 1
2 2
3 3
4 4
5 5

Title
1 Dynamic Backtracking
2 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
3 An Empirical Analysis of Search in GSAT
4 The Difficulties of Learning Logic Programs with Cut
5 Software Agents: Completing Patterns and Constructing User Interfaces

Category
1 ['cs.AI']
2 ['cs.AI']
3 ['cs.AI']
4 ['cs.AI']
5 ['cs.AI']

>
```

Environment

Object	Size	Variables
clean_exact	5 obs. of 3 variables	
combined_data	10000 obs. of 2 variables	
df	10000 obs. of 15 variables	
df_clean	10000 obs. of 4 variables	
df_part1	5000 obs. of 2 variables	
df_part2	5000 obs. of 2 variables	
duplicates_df	7 obs. of 3 variables	
duplicates_report	2 obs. of 4 variables	
job_title	3000 obs. of 18 variables	
long_df	20000 obs. of 4 variables	
unique_titles	5 obs. of 3 variables	

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prac 6.7.8.9&10 r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
anlv_al.csv	13.3 MB	Dec 8, 2025, 10:58 AM

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```
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Go to File/Function Addins
Source
R 4.5.2 - ~/RPROG/
> # =====
> # R Script: Vertical Concatenation using rbind()
> # Dataset: arxiv_ai.csv (Split into 2 parts)
> # =====
> library(dplyr)
> # =====
> # 1. SETUP: Import Data
> # =====
> df <- read.csv("arxiv_ai.csv", na.strings = c("", "NA"))
> print("---- Data Structure Before Transformation ----")
[1] "---- Data Structure Before Transformation ----"
> print(names(df)) # Show original column names
[1] "authors" "categories" "comment" "doi" "entry_id"
[6] "journal_ref" "pdf_url" "primary_category" "published" "summary"
[11] "title" "updated"
> # We will use two numeric columns for demonstrations:
> # - Year (from 'published')
> # - TitleLength (from 'title')
>
> df <- df %>%
+ mutate(
+   Species = categories, # Using 'categories' as Species
+   Height = nchar(title), # Using title length as Height
+   Year = as.numeric(substr(published, 1, 4))
+ )
> # Print preview
> print("---- Preview of Updated Dataset ----")
[1] "---- Preview of Updated Dataset ----"
> print(head(df[, c("Species", "Height", "Year")]))
  Species Height Year
1 [cs.AI]    20 1993
2 [cs.AI]   105 1993
3 [cs.AI]    39 1993
4 [cs.AI]    52 1993
5 [cs.AI]    69 1993
6 [cs.AI]    70 1993
>
> # =====
> # 2. DATA PREPARATION (Creating Two Compatible Datasets)
> # =====
> # rbind() will FAIL if columns do not match.
> # We will create two subsets with identical column structure:
```

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```
RPROG - RStudio
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Source
R 4.5.2 - ~/RPROG/
> # 2. DATA PREPARATION (Creating Two Compatible Datasets)
> #
> # rbind() will FAIL if columns do not match.
> # We will create two subsets with identical column structure:
> # Common Columns: "Species" and "Height"
>
> # 2.1 First half of the dataset
> df_part1 <- df[1:floor(nrow(df)/2), c("Species", "Height")]
> names(df_part1) <- c("Species", "Height")
>
> # 2.2 Second half of the dataset
> df_part2 <- df[(floor(nrow(df)/2)+1):nrow(df), c("Species", "Height")]
> names(df_part2) <- c("Species", "Height")
>
> # Ensure both Height columns are numeric
> df_part1$Height <- as.numeric(df_part1$Height)
> df_part2$Height <- as.numeric(df_part2$Height)
>
> # 3. VERTICAL COMBINATION (rbind)
>
> combined_data <- rbind(df_part1, df_part2)
>
> print("---- Combined Data Summary ----")
> print(paste("Part 1 rows:", nrow(df_part1)))
> print(paste("Part 2 rows:", nrow(df_part2)))
> print(paste("Total rows (Expected):", nrow(df_part1) + nrow(df_part2)))
> print(paste("Total rows (Expected): 10000"))
> print(paste("Total rows (Actual):", nrow(combined_data)))
> print(paste("Total rows (Actual): 10000"))
>
> print("---- Preview of Combined Data (Top and Bottom) ----")
> print(head(combined_data)) # First few rows from part 1
  Species Height
1 ['cs.AI']    20
2 ['cs.AI']   105
3 ['cs.AI']    39
4 ['cs.AI']    52
5 ['cs.AI']    69
6 ['cs.AI']    70
> print(tail(combined_data)) # Last few rows from part 2
9995
      Species Height
9995 ['cs.AI', 'cs.LG'] 116
```

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```
R452 - RPROG
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Source
> # 2.1 First half of the dataset
> df_part1 <- df[1:floor(nrow(df)/2), c("Species", "Height")]
> names(df_part1) <- c("Species", "Height")
>
> # 2.2 Second half of the dataset
> df_part2 <- df[(floor(nrow(df)/2)+1):nrow(df), c("Species", "Height")]
> names(df_part2) <- c("Species", "Height")
>
> # Ensure both Height columns are numeric
> df_part1$Height <- as.numeric(df_part1$Height)
> df_part2$Height <- as.numeric(df_part2$Height)
>
> # 3. VERTICAL COMBINATION (rbind)
>
> combined_data <- rbind(df_part1, df_part2)
>
> print("---- Combined Data Summary ----")
[1] "---- Combined Data Summary ----"
> print(paste("Part 1 rows:", nrow(df_part1)))
[1] "Part 1 rows: 5000"
> print(paste("Part 2 rows:", nrow(df_part2)))
[1] "Part 2 rows: 5000"
> print(paste("Total rows (Expected):", nrow(df_part1) + nrow(df_part2)))
[1] "Total rows (Expected): 10000"
> print(paste("Total rows (Actual):", nrow(combined_data)))
[1] "Total rows (Actual): 10000"
>
> print("---- Preview of Combined Data (Top and Bottom) ----")
[1] "---- Preview of Combined Data (Top and Bottom) ----"
> print(head(combined_data)) # First few rows from part 1
  Species Height
1 ['cs.AI']    20
2 ['cs.AI']   105
3 ['cs.AI']    39
4 ['cs.AI']    52
5 ['cs.AI']    69
6 ['cs.AI']    70
> print(tail(combined_data)) # Last few rows from part 2
      Species Height
9995 ['cs.AI', 'cs.L0'] 116
9996 ['cs.CL', 'cs.AI'] 102
9997 ['q-fin.RM', 'cs.AI'] 49
9998 ['cs.NI', 'cs.AI'] 78
9999 ['cs.L0', 'cs.AI'] 63
10000 ['cs.AI', 'cs.L0', '03860, 03815, 68T27, 68T30, 68T15', 'I.2.3; I.2.4; I.2.0; F.4.1'] 50
>
```

Environment History Connections Git Tutorial

Data

Object	Size	Obs.	Vars.
arkiv_ai	10000 obs.	12	vars.
category_pivot	10000 obs.	919	vars.
combined_data	10000 obs.	2	vars.
df	10000 obs.	15	vars.
df_clean	10000 obs.	4	vars.
df_part1	5000 obs.	2	vars.
df_part2	5000 obs.	2	vars.
job_title	3000 obs.	18	vars.
long_df	20000 obs.	4	vars.
wide_df	10000 obs.	4	vars.

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

Name	Size	Modified
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analysis.R		
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prac.67.8.9&prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac.pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
arkiv_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

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